

Build and count cubic units

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: A solid has 5 layers of 24 unit cubes. What is its volume? Copy or complete the first step.

2. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

3. A cuboid has volume 360 cm^3 , length 12 cm and width 5 cm. Find its height.

4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A solid has 5 layers of 24 unit cubes. What is its volume? Copy or complete the first step.
Answer 120 cubic units.
2. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
Answer 420 cm³.
3. A cuboid has volume 360 cm³, length 12 cm and width 5 cm. Find its height.
Answer 6 cm.
4. Miro says: Miro calculates the surface area or adds the three dimensions. Is this correct? Explain.
Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.

Build and count cubic units

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A solid has 5 layers of 24 unit cubes. What is its volume?

6. Check this result using an inverse, estimate or known fact: A cuboid has volume 360 cm^3 , length 12 cm and width 5 cm. Find its height.

2. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

7. Prove or explain your reasoning: Cuboid A is $8 \times 6 \times 5 \text{ cm}$. Cuboid B is $10 \times 6 \times 4 \text{ cm}$. Compare their volumes.

3. A cuboid has volume 360 cm^3 , length 12 cm and width 5 cm. Find its height.

8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.

4. Solve this independently and show every stage: A solid has 5 layers of 24 unit cubes. What is its volume?

9. Write a complete sentence explaining how you checked one answer.

5. Use a second method or representation for: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

10. Write and solve a similar question to: A cuboid has volume 360 cm^3 , length 12 cm and width 5 cm. Find its height.

Teacher answers and checking prompts

1. A solid has 5 layers of 24 unit cubes. What is its volume?
Answer 120 cubic units.
2. Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
Answer 420 cm³.
3. A cuboid has volume 360 cm³, length 12 cm and width 5 cm. Find its height.
Answer 6 cm.
4. Solve this independently and show every stage: A solid has 5 layers of 24 unit cubes. What is its volume?
Answer 120 cubic units.
5. Use a second method or representation for: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
Answer Expected result: 420 cm³. Method or representation may vary.
6. Check this result using an inverse, estimate or known fact: A cuboid has volume 360 cm³, length 12 cm and width 5 cm. Find its height.
Answer Expected result: 6 cm. A valid check must be shown.
7. Prove or explain your reasoning: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
Answer Both are 240 cm³. The explanation must show why the method works.
8. Identify and correct this misconception: Miro calculates the surface area or adds the three dimensions.
Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.
9. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.
10. Write and solve a similar question to: A cuboid has volume 360 cm³, length 12 cm and width 5 cm. Find its height.
Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Build and count cubic units

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Prove or explain your reasoning: Cuboid A is $8 \times 6 \times 5$ cm. Cuboid B is $10 \times 6 \times 4$ cm. Compare their volumes.

6. Create a new example that tests the same idea as: A cuboid has volume 360 cm^3 , length 12 cm and width 5 cm. Find its height.

2. Prove the answer to this question, rather than only calculating it: A solid has 5 layers of 24 unit cubes. What is its volume?

7. Find a second method or representation. Compare its efficiency with your first method.

3. Solve this in two different ways and compare them: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.

8. Write one always, sometimes or never statement about today's learning and justify it.

4. Work backwards from the answer to reconstruct the question: 6 cm.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Cuboid A is 8 x 6 x 5 cm. Cuboid B is 10 x 6 x 4 cm. Compare their volumes.
Answer Both are 240 cm³. The explanation must show why the method works.
2. Prove the answer to this question, rather than only calculating it: A solid has 5 layers of 24 unit cubes. What is its volume?
Answer Expected result: 120 cubic units. Proof or justification must match the lesson structure.
3. Solve this in two different ways and compare them: Find the volume of a cuboid 12 cm by 7 cm by 5 cm.
Answer Expected result: 420 cm³. Two valid methods and a comparison are required.
4. Work backwards from the answer to reconstruct the question: 6 cm.
Answer One valid reconstruction is: A cuboid has volume 360 cm³, length 12 cm and width 5 cm. Find its height. Other equivalent questions are acceptable.
5. Explain why this reasoning fails: Miro calculates the surface area or adds the three dimensions.
Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.
6. Create a new example that tests the same idea as: A cuboid has volume 360 cm³, length 12 cm and width 5 cm. Find its height.
Answer Answers vary. The example and solution must preserve the mathematical structure.
7. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
8. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.
9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
Answer Answers vary. The error must be mathematically relevant and the correction must be precise.
10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.